Department of Information Technology

Academic Year 2024 - 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT
Course Code & Title: CS3352 & Foundations of Data Science
Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / **Basic statistical Description of data**
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO2
- **Activity Chosen:** One Minute Paper
- **Justification:**
  In unit I, Basic statistical description of data is an important topic needed to perform data analysis because the data analysis done by using statistics. This activity helps the students to know the basic concepts of statistics.
- **Time Allotted for the Activity:** 5 Minutes
- **Details of the Implementation:**
  - The topic Basic statistical Description of data was taught in the previous class.
  - Mean, Median, Mode explained with an example for grouped data and ungrouped data.
  - The time allotted for the topic is 5 Minutes.
  - The students were asked to write about mean, median and mode to assess the understanding level as shown in Figure 1 & 2.

- **CO – PO / PSO mapping:**

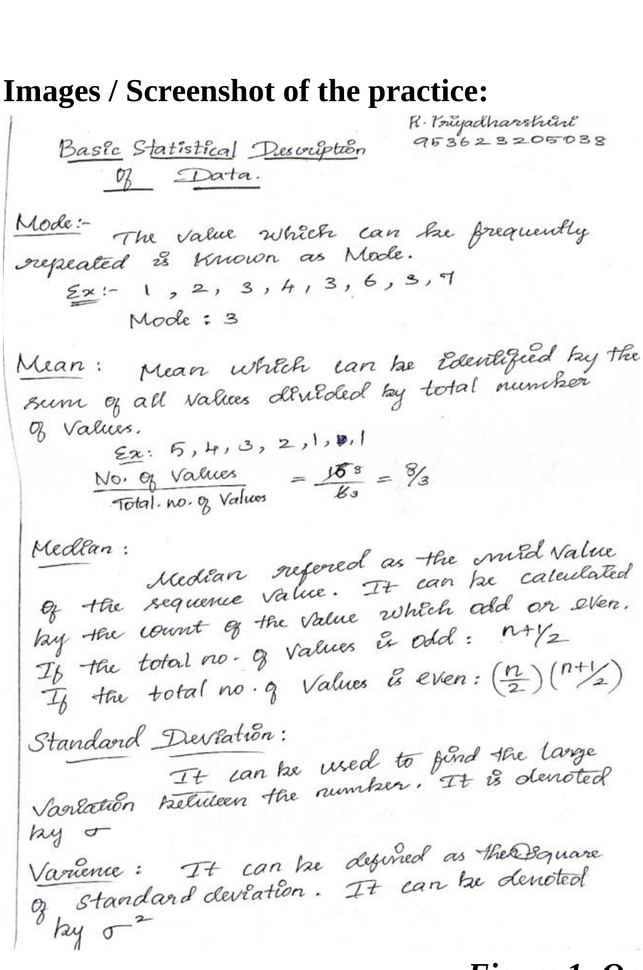
CO	PO1	PO2	PO3	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	2	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO12**	**PSO1**	**PSO2**	**PSO3**
	3	3	2	2	2	3	2
Justification for correlation	Apply basic Knowledge and fundamentals of mathematics for statistical description on data	The students will be able to identify the need for analysis of complex engineering problems	Able to design the process flow for developing data science applications.	The students able to recognize the need for statistics for data science process.	The students able to exhibit the knowledge in statistics	The students can Exhibit knowledge of data science in the field of IoT, Cloud Computing to develop IT solutions	The students able to exhibit knowledge in statistics develop a quality product.

- **Images / Screenshot of the practice:**



R. Priyadharshini
953623205038

Basic Statistical Description of Data.

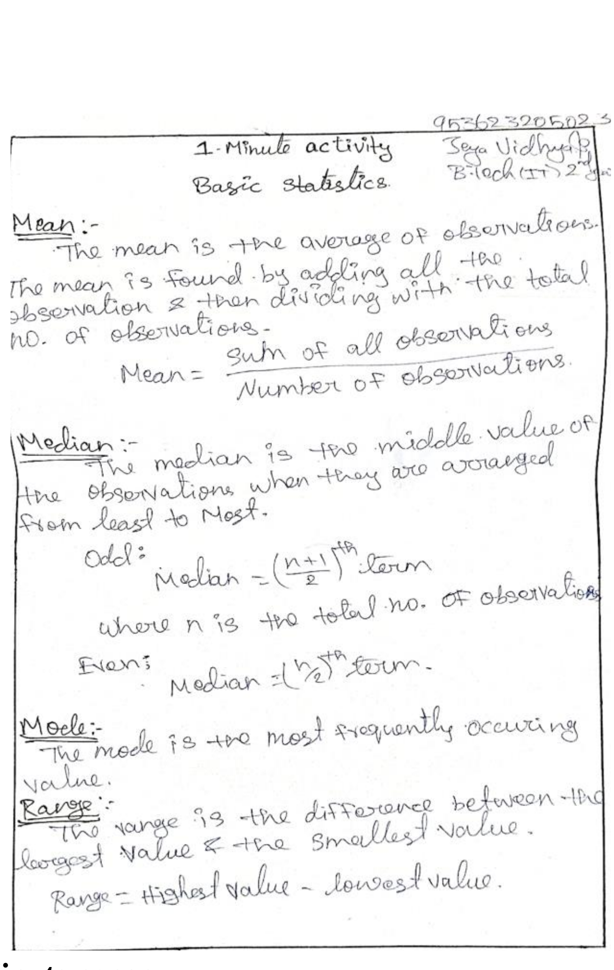
Mode:- The value which can be frequently repeated is known as Mode.
Ex:- 1, 2, 3, 4, 3, 6, 3, 7
Mode : 3

Mean: Mean which can be Identified by the sum of all values divided by total number of values.
Ex: 5, 4, 3, 2, 1, 1
$\frac{\text{No. of Values}}{\text{Total no. of Values}} = \frac{16}{6} = \frac{8}{3}$

Median: Median refered as the mid value of the sequence value. It can be calculated by the count of the value which odd or even.
If the total no. of values is odd: n+1/2
If the total no. of values is even: $\left(\frac{n}{2}\right)\left(\frac{n+1}{2}\right)$

Standard Deviation: It can be used to find the large variation between the number. It is denoted by σ

Varience: It can be defined as the square of standard deviation. It can be denoted by σ^2



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1-Minute activity
Jeya Vidhya
B.Tech (IT) 2nd yr
Basic statistics.

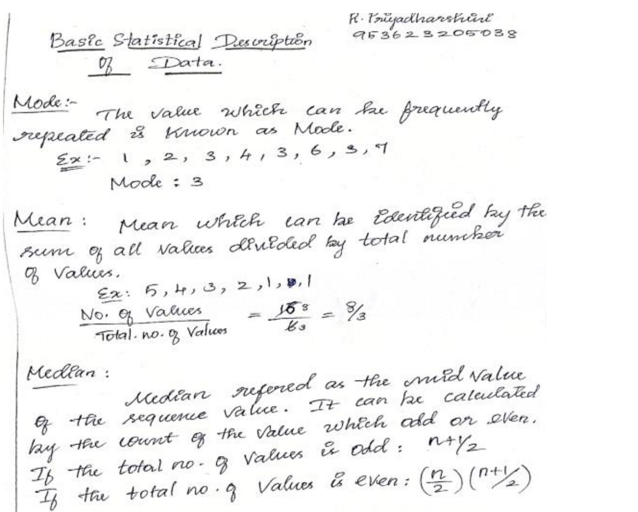
Mean:- The mean is the average of observations. The mean is found by adding all the total observation & then dividing with the total no. of observations.
Mean = $\frac{\text{Sum of all observations}}{\text{Number of observations}}$

Median:- The median is the middle value of the observations when they are arranged from least to Most.
Odd: Median = $\left(\frac{n+1}{2}\right)^{th}$ term
where n is the total no. of observations
Even: Median = $\left(\frac{n}{2}\right)^{th}$ term.

Mode:- The mode is the most frequently occuring value.

Range:- The range is the difference between the largest value & the smallest value.
Range = Highest value - lowest value.

Figure 1: One Minute paper



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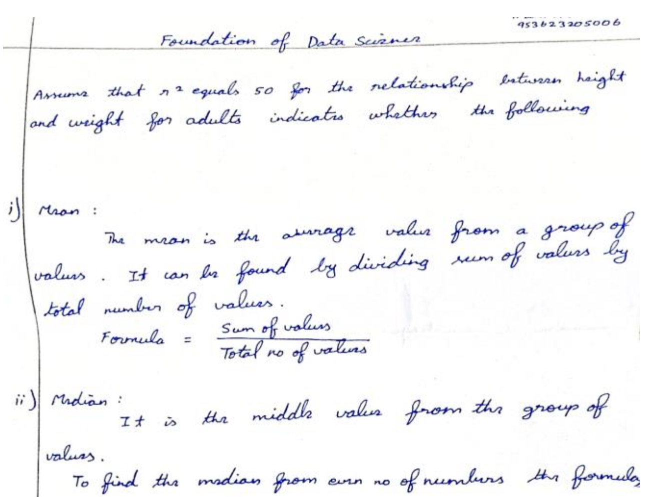

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Foundation of Data Science

Assume that r^2 equals 50 for the relationship between height and weight for adults indicates whether the following

i) Mean:
The mean is the average value from a group of values. It can be found by dividing sum of values by total number of values.
Formula = $\frac{\text{Sum of values}}{\text{Total no of values}}$

ii) Median:
It is the middle value from the group of values.
To find the median from even no of numbers the formula
$= \frac{n+1}{2}$
To find the median from the odd no. of numbers, the formula is =

iii) Mode.
It is the highest repeated numbers

Figure 2 One Minute Paper Activity

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  - Students said that they were able to understand the concepts of Statistical description of data
  - The students said that the activity help to recall the concepts.
  - It makes the students to improve their thinking skills.
- ***Benefit of the practice:***
  - Students can able to recall the different statistical methods and their applications in data science process.
  - The students can practice statistical methods with examples.
- ***Challenges faced in implementation:***
  - The Activity planned for 5 minutes but it takes 10 minutes to complete the activity.
  - Some students not able to recall the concepts.

References:

- https://oncourseworkshop.com/self-awareness/one-minute-paper/
- https://www.ijfcm.org/article-details/13601

Signature of Faculty Member **HOD**